CYSC H1001 Network Security

Open-Source Firewall Project

An investigation into OPNsense, IPcop and IPFire.



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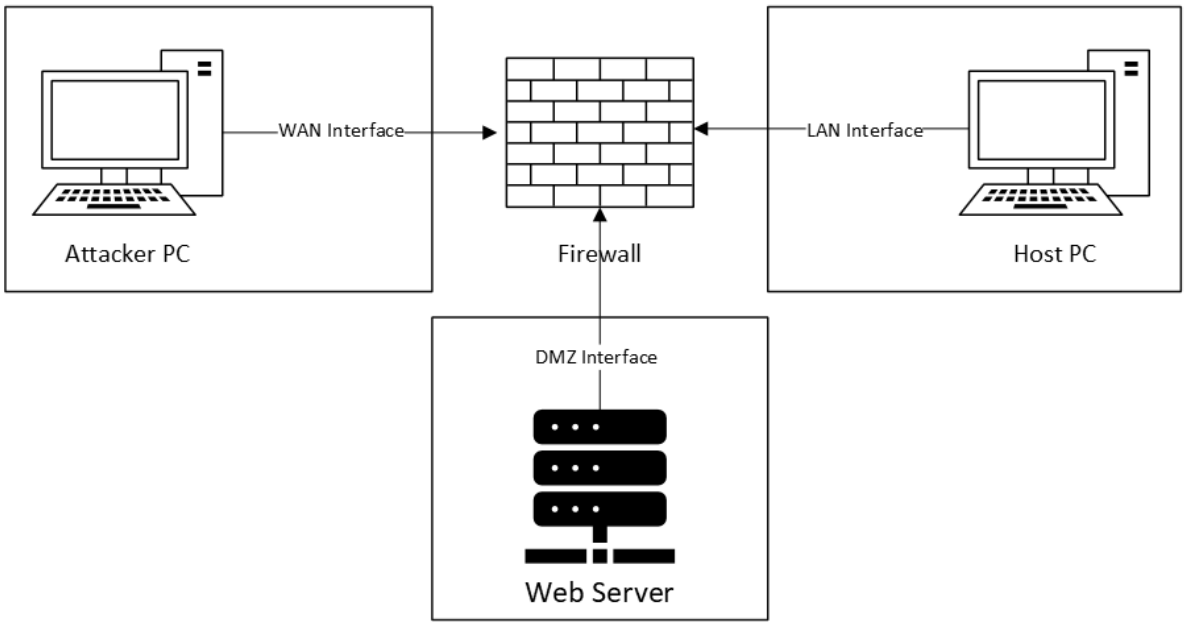
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Background

Topology



The topology consists of three networks each connected to an interface on the firewall. An outside network connected to a WAN (NAT to internet), a DMZ with a web server (port forwarded), and an Internal network for all other devices.

The attacker PC will run Kali Linux with the penetration testing tools that are required for the attacks we are trying to mitigate, such as hping3, Nmap and Hydra.

All firewalls will initially be configured with a base configuration which will allow internet access for all networks and communication between subnets. The attacks will be run before and after applying the necessary rules and enabling/disabling certain features which should mitigate the attacks.

Tests overview

We will focus on attacks from the WAN side, so no physical access to the firewall and its network is required. The attacks that we will be trying to mitigate via the individual firewalls are:

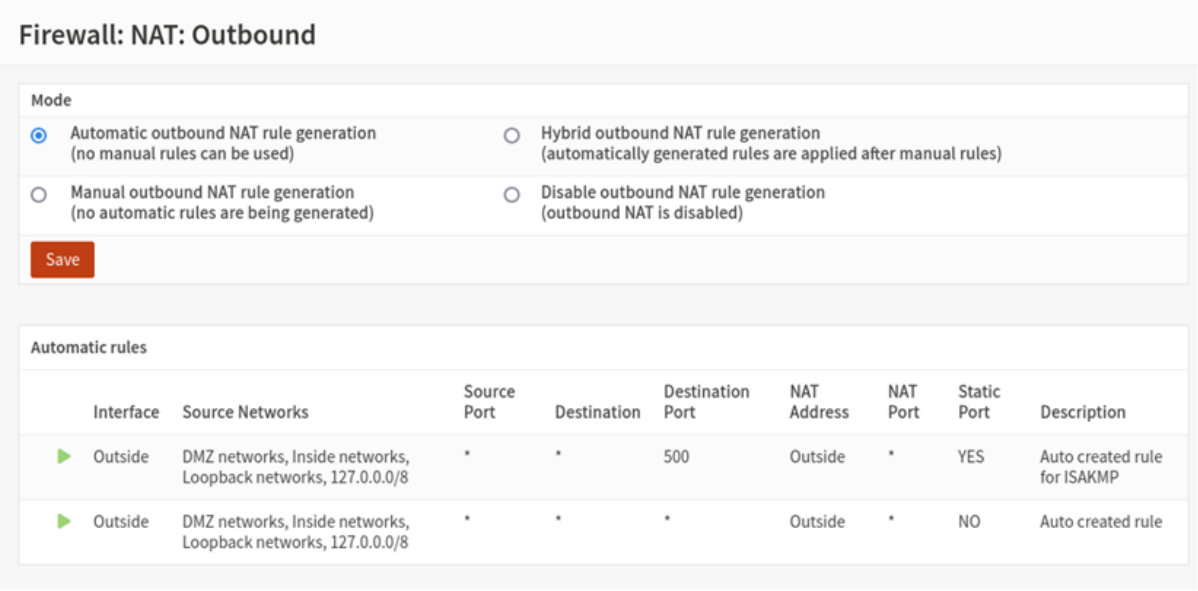
- **Reconnaissance scans/attacks,** where a threat actor uses a tool such as Nmap to gather information about a network and identify any potential vulnerabilities that they can exploit (MITRE ATT&CK, 2020).
    - We will use Nmap scans and pings to test this.
- Denial of service attacks, such as **ICMP Floods** and **HTTP Floods**, where excess ICMP or HTTP packets are sent to a host. (Cloudflare, 2024)
    - We will use hping3 and wrk to perform these attacks.
- Attacks on the firewalls themselves We will focus on **SSH**, because this is the most likely form of remote management.
    - We will simply attempt logins via SSH and use hydra to test brute forcing.

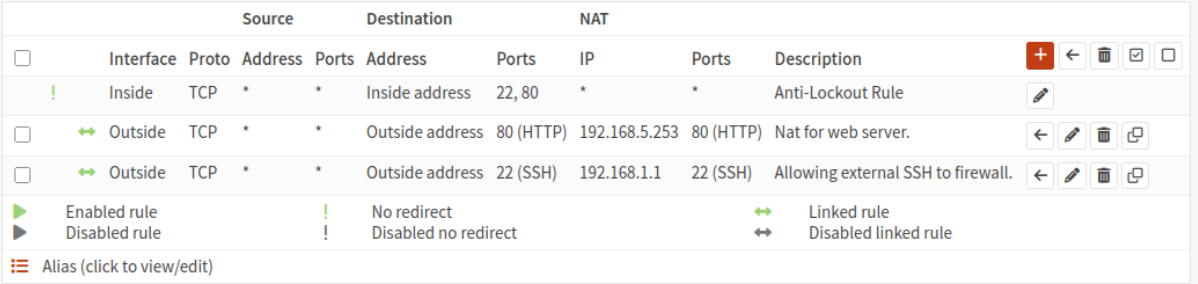
OPNsense (Adam Lamont)

OPNsense is an open source Layer 3 & 4 firewall that is a fork of pfSense, it is based on FreeBSD, unlike the other firewalls in this report which are based on Linux. It comes with packet filter, IPS/IDS using Suricata, built in DDOS protection and traffic normalization (OPNsense, 2025). These are all areas of interest in relation to the attacks that we are trying to mitigate.

Summary of default configuration

- Enabled port forward NAT for web server, so port 80 on wan interface points to web server.
  Enabled ssh on firewall, port 22 on firewall points to ssh (internally and externally)
- Created basic allow rules for http and ssh traffic only.
- Verified traffic can flow from inside to outside, DMZ to outside, outside into DMZ, but not inside.





Attacks before firewall configuration

Configuration of packet filter

Firewall: Rules: DMZ

Select category | Inspect

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
								Automatically generated rules (9)
	IPv4 *	DMZ net	*	*	*	*	*	Gives DMZ internet access.
	IPv4 TCP	*	*	192.168.5.253	80 (HTTP)	*	*	Allowing http traffic into web server.
	IPv4 *	DMZ net	*	*	*	*	*	Default allow LAN to any rule

pass | pass (disabled) | block | block (disabled) | reject | reject (disabled) | log | log (disabled) | in | out | first match | last match

Active/Inactive Schedule (click to view/edit)

Alias (click to view/edit)

DMZ rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.

Firewall: Rules: Inside

Select category | Inspect

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
								Automatically generated rules (19)
	IPv4 *	Inside net	*	*	*	*	*	Default allow LAN to any rule
	IPv6 *	Inside net	*	*	*	*	*	Default allow LAN IPv6 to any rule

pass | pass (disabled) | block | block (disabled) | reject | reject (disabled) | log | log (disabled) | in | out | first match | last match

Active/Inactive Schedule (click to view/edit)

Alias (click to view/edit)

Inside rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.

Firewall: Rules: Outside

Select category | Inspect

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
								Automatically generated rules (12)
	IPv4 TCP	*	*	192.168.5.253	80 (HTTP)	*	*	Web Server allowing

pass | pass (disabled) | block | block (disabled) | reject | reject (disabled) | log | log (disabled) | in | out | first match | last match

Active/Inactive Schedule (click to view/edit)

Alias (click to view/edit)

Outside rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.

- All tests are performed WAN side, assuming attacker has no internal access.

Reconnaissance Attacks

Plain pings are unsuccessful.

```
└─$ ping 192.168.183.243
PING 192.168.183.243 (192.168.183.243) 56(84) bytes of data.
^C
--- 192.168.183.243 ping statistics ---
10 packets transmitted, 0 received, 100% packet loss, time 9217ms
```

Also immune to fragmented hping3 ping:

```
└─$ sudo hping3 -1 -f -d 1500 192.168.183.243
HPING 192.168.183.243 (eth0 192.168.183.243): icmp mode set, 28 headers + 150
0 data bytes
^C
── 192.168.183.243 hping statistic ──
17 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

Nmap scan (OS detection and all ports):

```
└─$ sudo nmap -A -p- 192.168.183.243
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-25 14:23 EDT
Nmap scan report for 192.168.183.243
Host is up (0.00078s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.0 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 00:0C:29:8A:18:49 (VMware)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): FreeBSD 11.X (92%)
OS CPE: cpe:/o:freebsd:freebsd:11.2
Aggressive OS guesses: FreeBSD 11.2-RELEASE (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: FreeBSD; CPE: cpe:/o:freebsd:freebsd

TRACEROUTE
HOP RTT     ADDRESS
1   0.78 ms 192.168.183.243

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 120.22 seconds
```

From this scan, the attacker can identify that the firewall runs on FreeBSD (so could be OPNsense/PFsense), and that the http server runs on Apache and is using Ubuntu, and ports 22 and 80 are open.

Denial of Service Attacks

As seen above, pinging the firewall with hping3 did not work. However, if we flood the firewall with fragmented ping packets, the firewall will still not reply, but if you floods the wan interface with packets, and this can be seen in the logs of the firewall. It could be used to perform a denial-of-service attack on it.

```
└─$ sudo hping3  -1 --flood -f -d 1500 192.168.183.243
```

20 seconds of pinging:

Outside	192.168.183.246	0.0 b	86.7 mb	0b	86.7 mb	0 Bytes	46 MB	2025-10-25T18:39:42.280Z

After a minute:

Outside	192.168.183.246	0.0 b	107.0 mb	0b	115.0 mb	0 Bytes	172 MB	2025-10-25T18:40:58.335Z

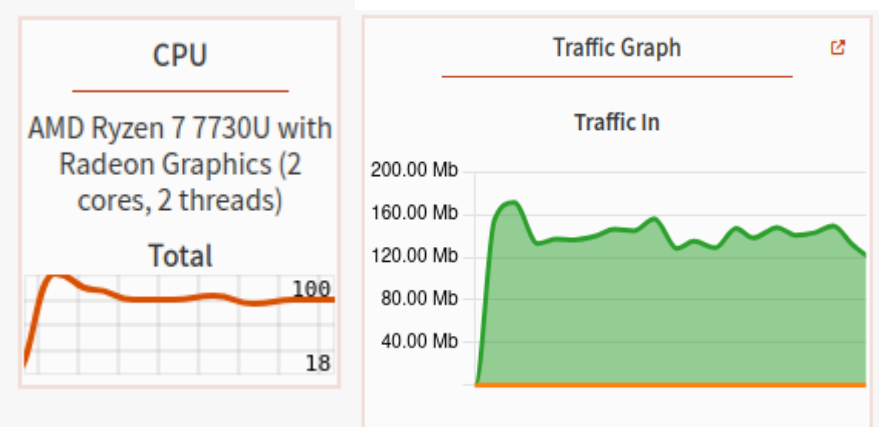
I then ran the attack with an option to use random source IP addresses, and this produced interesting results.

```
└─$ sudo hping3  -1 --flood --rand-source -x -d 1500 192.168.183.243
```

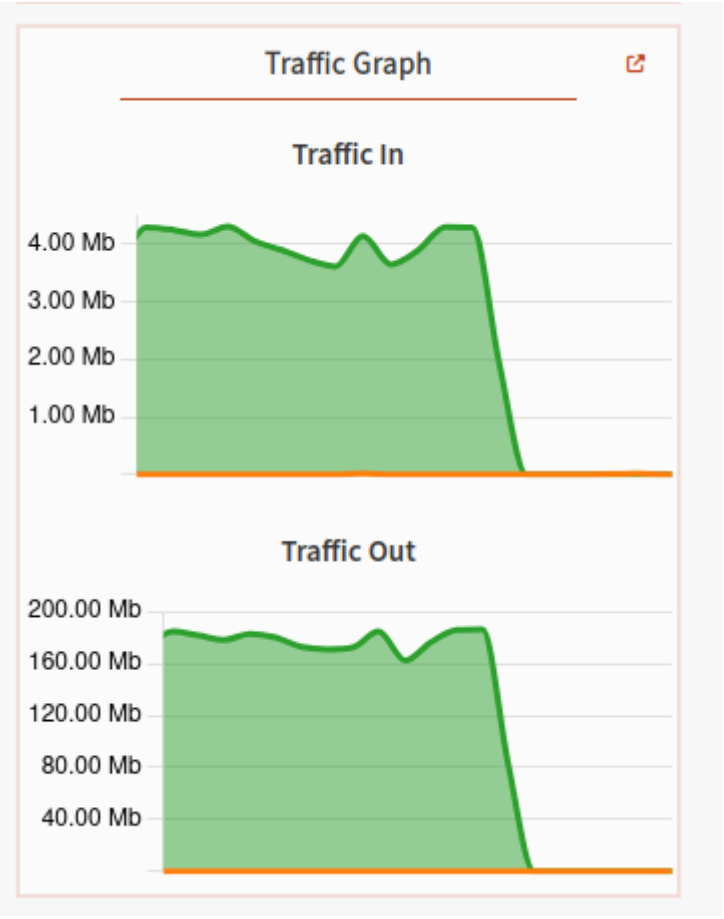
It produced an error on the cli,

```
[zone: pf frag entries] PF frag entries limit reached
```

but while running this attack and accessing the web server with another host, it had no issue accessing it. The CPU of the firewall is being put under lots of load, and in a real-world scenario this will cause problems. The fact this attack is possible is a security issue, because it generates a huge amount of incoming traffic on the WAN interface.



Using a tool called wrk, a DOS can be performed using http traffic.




```
└─$ wrk -t 4 -c 4 -d 20 http://192.168.183.243
Running 20s test @ http://192.168.183.243
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency     2.01ms  544.32us  11.97ms   79.84%
    Req/Sec   498.17     47.39   650.00     82.50%
  39730 requests in 20.04s, 413.99MB read
Requests/sec:   1982.37
Transfer/sec:     20.66MB
```

As can be seen, 413.99MB of traffic can be generated by one host in 20 seconds. This is problematic if it was performed in a distributed manner.

SSH Attacks

The web-based management of the firewall is not publicly accessible, but ssh is however, and this can be exploited. As an attacker can enumerate the type of firewall and web server used, they can know the default login for the firewall is root.

```
└─$ ssh root@192.168.183.243
(root@192.168.183.243) Password:
(root@192.168.183.243) Password:
(root@192.168.183.243) Password:

root@192.168.183.243's password:
Permission denied, please try again.
root@192.168.183.243's password:
Permission denied, please try again.
root@192.168.183.243's password:
Received disconnect from 192.168.183.243 port 22:2: Too many authentication f
ailures
Disconnected from 192.168.183.243 port 22
```

There is a lockout configured by default, however.

Using hydra to crack the password:

```
hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt ssh://192.168.183.243:22
```

However, again the timeout kicks in and the attempts fail.

Date	Severity	Process	Line	
2025-10-25T19:12:26	Error	sshd-session	error: maximum authentication attempts exceeded for root from 192.168.183.246 port 56252 ssh2 [preauth]	→
2025-10-25T19:12:26	Error	sshd-session	error: maximum authentication attempts exceeded for root from 192.168.183.246 port 56254 ssh2 [preauth]	→
2025-10-25T19:12:26	Error	sshd-session	error: maximum authentication attempts exceeded for root from 192.168.183.246 port 56250 ssh2 [preauth]	→
2025-10-25T19:12:25	Warning	audit	user root could not authenticate for sshd. [using OPNsense\Auth\Services\System + OPNsense\Auth\Local]	→
2025-10-25T19:12:25	Warning	audit	user root could not authenticate for sshd. [using OPNsense\Auth\Services\System + OPNsense\Auth\Local]	→
2025-10-25T19:12:25	Warning	audit	user root could not authenticate for sshd. [using OPNsense\Auth\Services\System + OPNsense\Auth\Local]	→
2025-10-25T19:12:22	Warning	audit	user root could not authenticate for sshd. [using OPNsense\Auth\Services\System + OPNsense\Auth\Local]	→
2025-10-25T19:12:22	Warning	audit	user root could not authenticate for sshd. [using OPNsense\Auth\Services\System +	→

« ‹ 1 2 ›

Showing 1 to 20

Timeout makes brute force take a long time:

```
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:143443
99), ~3586100 tries per task
[DATA] attacking ssh://192.168.183.243:22/
[STATUS] 40.00 tries/min, 40 tries in 00:01h, 14344359 to do in 5976:49h, 4 active
[STATUS] 28.00 tries/min, 84 tries in 00:03h, 14344315 to do in 8538:17h, 4 active
[STATUS] 26.29 tries/min, 184 tries in 00:07h, 14344215 to do in 9095:04h, 4 active
```

Configuration that was applied

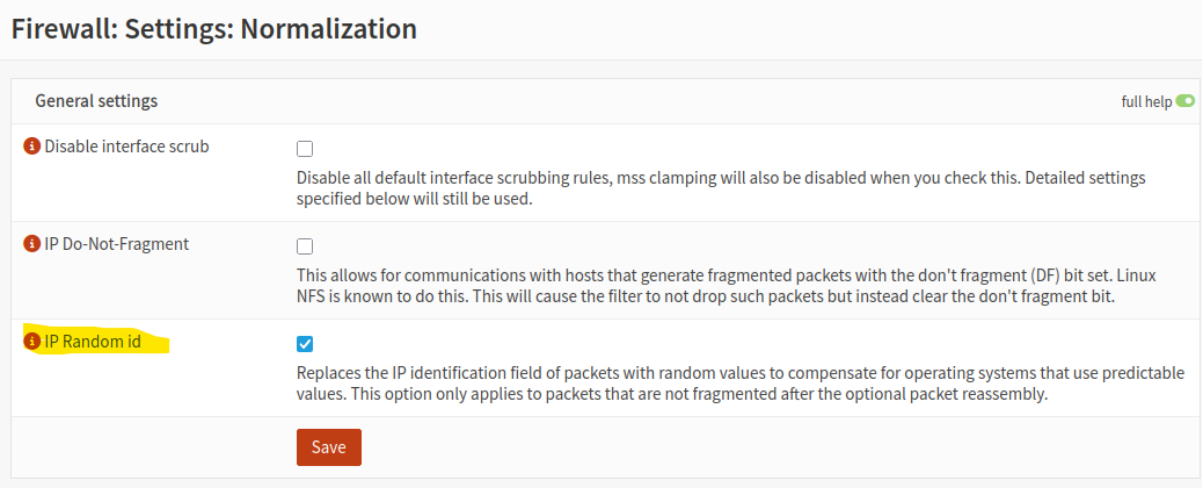
- Rate-limiting I/O from a single IP with specific protocols (HTTP) using shaper function.
- Tighter access control for SSH (Limiting to certain IP ranges) through rules.
- Blocking ICMP Echo Requests, unless originating from LAN side through rules
- Blocking malicious IP ranges by default using aliases (groups of IP addresses and ranges).

Attacks after firewall configuration

Reconnaissance Attacks

Web server IP being revealed is to do with web server itself, so it is outside the scope of this project.

Enabling this IP Random ID setting makes the Nmap os scan less verbose, but only slightly.




```
└─$ nmap -A -p- 192.168.183.243
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-26 15:46 EDT
Nmap scan report for 192.168.183.243
Host is up (0.025s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
51311/tcp open  ssh     OpenSSH 10.0 (protocol 2.0)
Service Info: OS: FreeBSD; CPE: cpe:/o:freebsd:freebsd

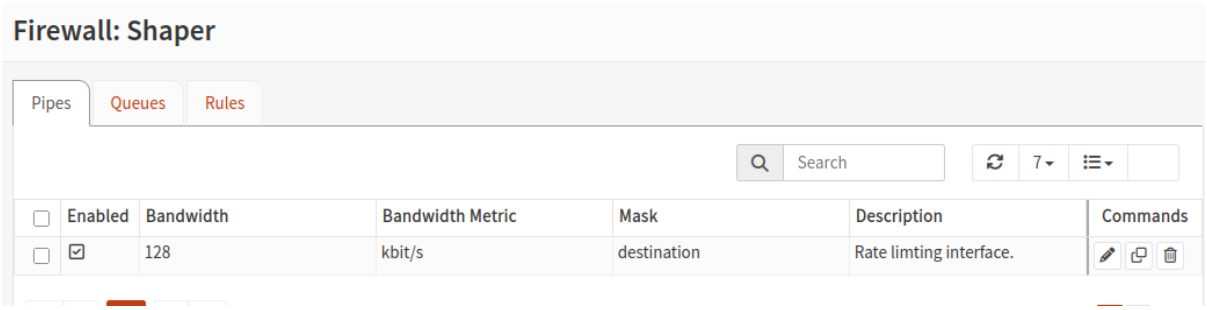
Service detection performed. Please report any incorrect results at https://nmap.org
/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 115.97 seconds
```

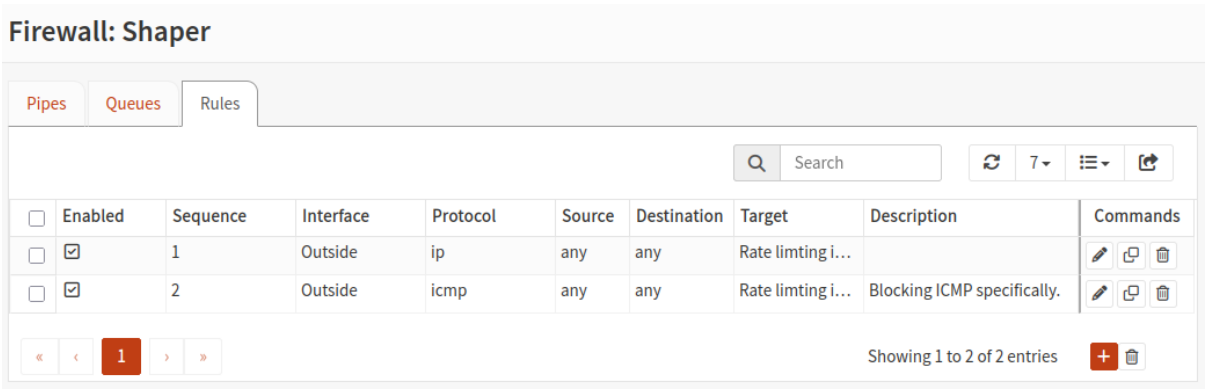
Dos Attacks

Reduces effectiveness of ICMP flood attack via floating rule to block ICMP echo requests on LAN interface. Has negative side effect of increasing CPU usage, due to logging.



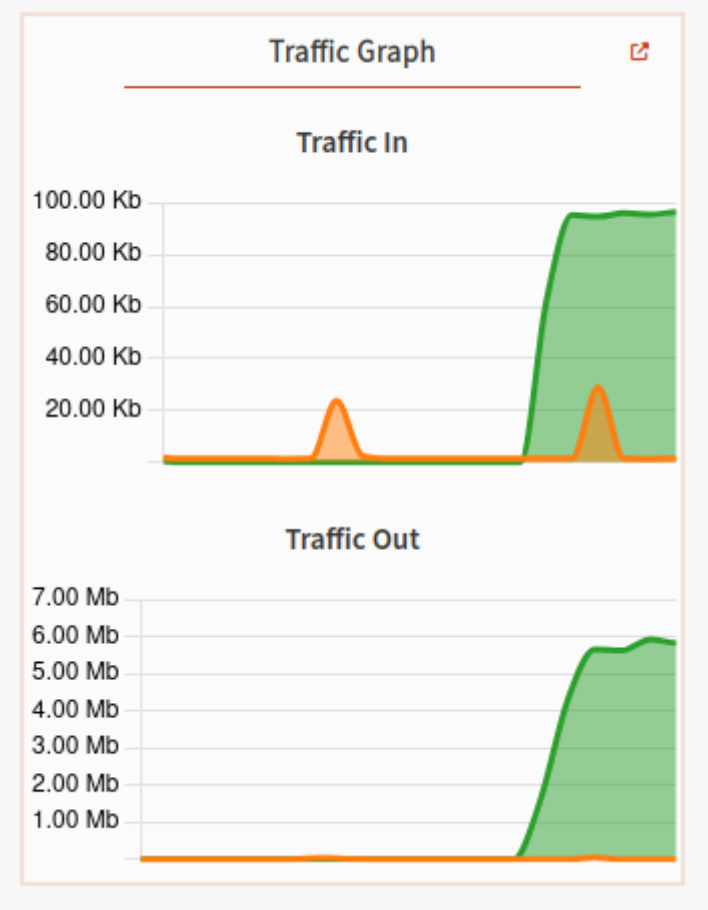
Shaper was used to rate limit http traffic.





(Rule 1, rule 2 was trying to block ICMP, but it failed to work.)

Wrk with shaper enabled. Much less traffic going in and out.



Firewall: Shaper: Status

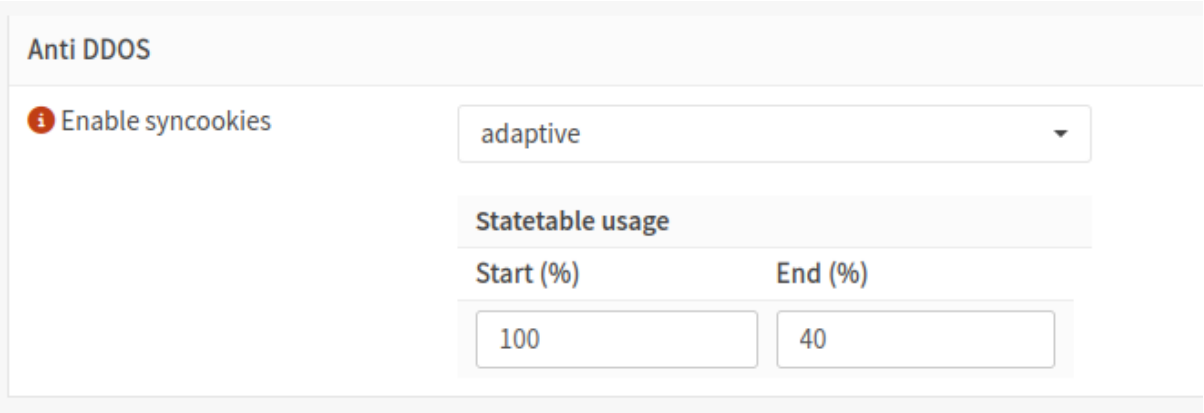
Current Activity

Show rules
Show active flows

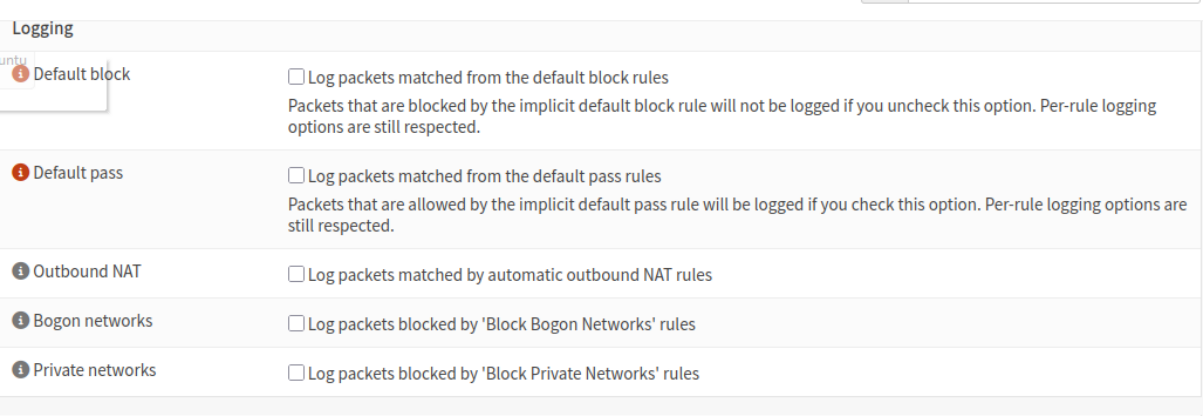
	#	Description	Bandwidth	Packets	Bytes	Accessed
☐	10001		10.000 Kbit/s			
+	10001.141073		0			
☐	10000	Rate limting interface.	80.000 Kbit/s	3.01k	203.98k	2025-10-27T20:36:31
+	10000.141072		0	3.01k [100.00 %]	203.98k [100.00 %]	2025-10-27T20:36:31

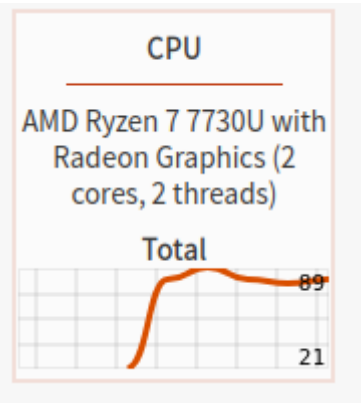
```
┌──(kali㉿kali)-[~]
└─$ wrk -t 4 -c 4 -d 20 http://192.168.183.243
Running 20s test @ http://192.168.183.243
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency    68.47ms   15.60ms 131.09ms   83.63%
    Req/Sec    14.59      5.16    20.00     49.87%
  1161 requests in 20.09s, 12.10MB read
Requests/sec:     57.78
Transfer/sec:    616.57KB
```

I also enabled this anti DDOS setting, didn't have a noticeable impact, as attacks were via ICMP.



Disabling logging of default rules seems to reduce CPU usage to high 70-80 instead of 100%. The packets are never being processed, just hammering the logs with blocks.





Despite high CPU usage, firewall is noticeably more responsive with these logs disabled This attack should be prevented at an ISP level, so it would be very difficult to perform in the real world. Increasing the CPU core count of the virtual machine from 2 to 4 further reduces this issue and causes maximum load of 50%. This could vary however, if this was done via a distributed botnet.

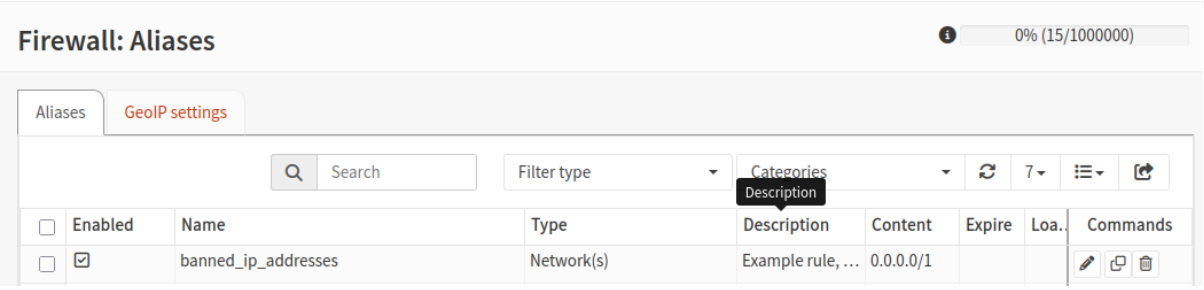
SSH Attacks

Rule applied:



Only allows traffic from specific IP addresses. (Address shown is host pc)

A list of IPs can be created as alias, e.g. used for blocking specific ranges. In real world, use GeoIP lists, and lists of malicious addresses. Example was created for half of all IP addresses, which did generate logs when testing spoofed IP address ICMP attack.



Port was changed from 22 to 51311. This is for security, will only be useful if attacker doesn't run a full Nmap scan on all ports.

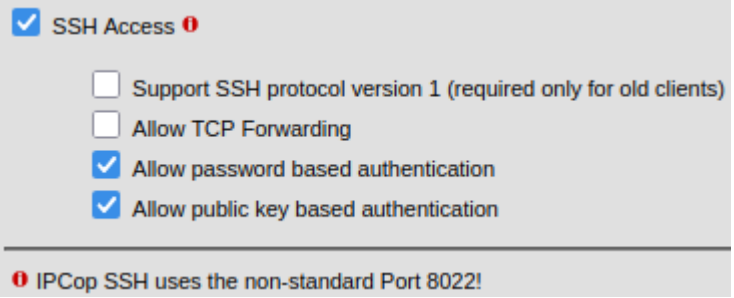
IPCop (Roisin Aylward)

Background

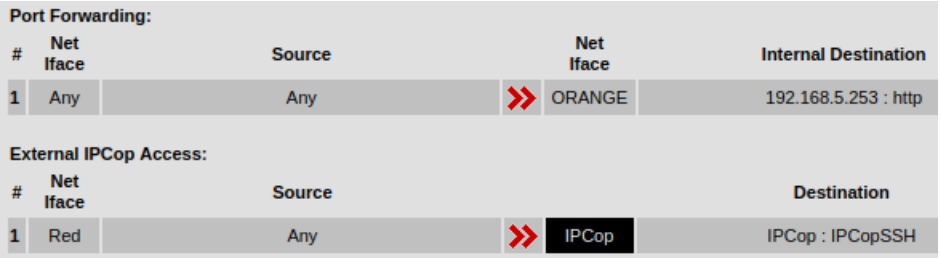
IPCop is an open-source Linux firewall that is a fork of Smoothwall. It is now considered obsolete with its last version 2.1.9 released in 2015 (IPCop, 2016). It aims to be user friendly by providing a web interface that simplifies tasks (Dempster & Eaton-Lee, 2006). IPCop has four network interfaces, each represented by colours. Green is used for the hosts in the internal network that IPCop is protecting, blue is for an optional semi-trusted wireless network, orange is used for servers in an optional DMZ network, and red is for the Internet or other untrusted network that IPCop aims to protect the Green, Blue and Orange networks from. (IPCop, 2016)

Initial Configuration

IPCop's web interface is being accessed by an Xubuntu virtual machine in the green network. SSH Access was enabled on IPCop, which uses the port 8022 instead of the standard port 22.



The firewall rules were configured to allow SSH connections to IPCop from the red network so that any device can remotely access the firewall terminal, and to port forward any HTTP traffic to the webserver.



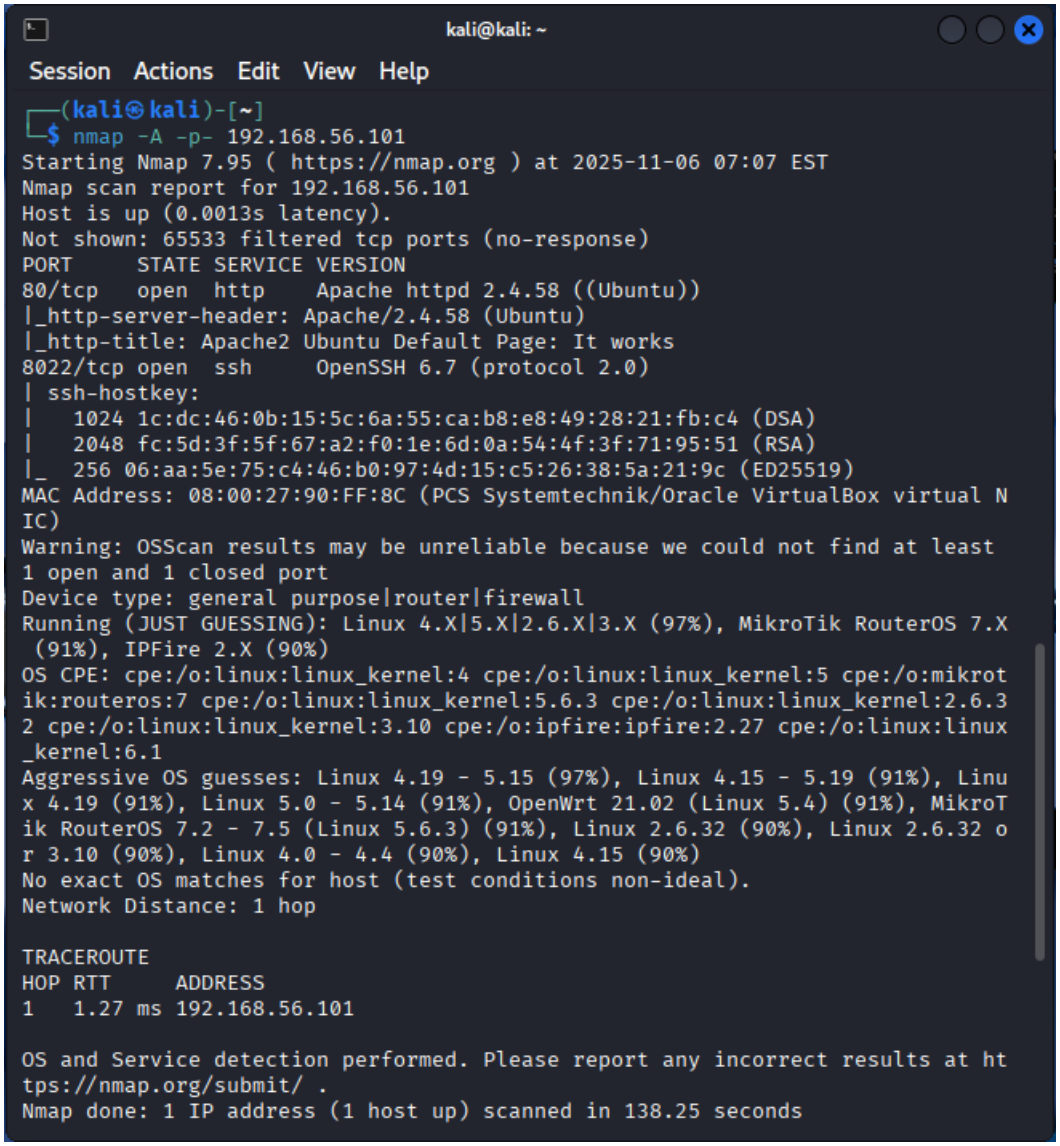
Attacks Before Reconfiguration

A Kali Linux virtual machine on the WAN network attempts to attack IPCop's red network IP address, 192.168.56.101.

Reconnaissance Attacks

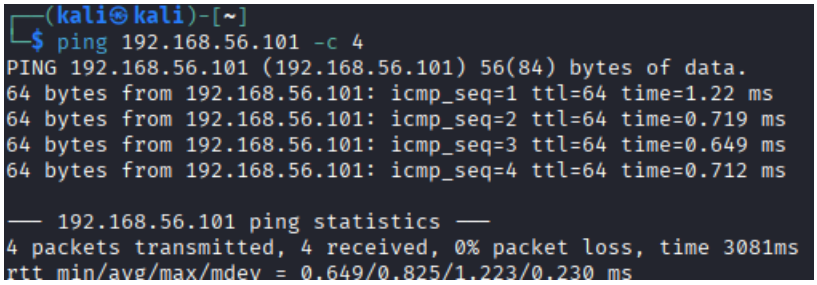
The reconnaissance attack is achieved using Nmap to scan all of the ports on the IP address, with Operating System detector and traceroute. The Nmap scan report reveals that two ports are open: port 80/TCP which is being used by a HTTP Apache2 server on Ubuntu, and port 8022/TCP which is an SSH protocol. The Operating System detector

correctly guesses that IPCop is a Linux which is being used as a firewall but incorrectly guesses IPFire as the distribution. The traceroute shows that the red network is one hop away from the attacker.

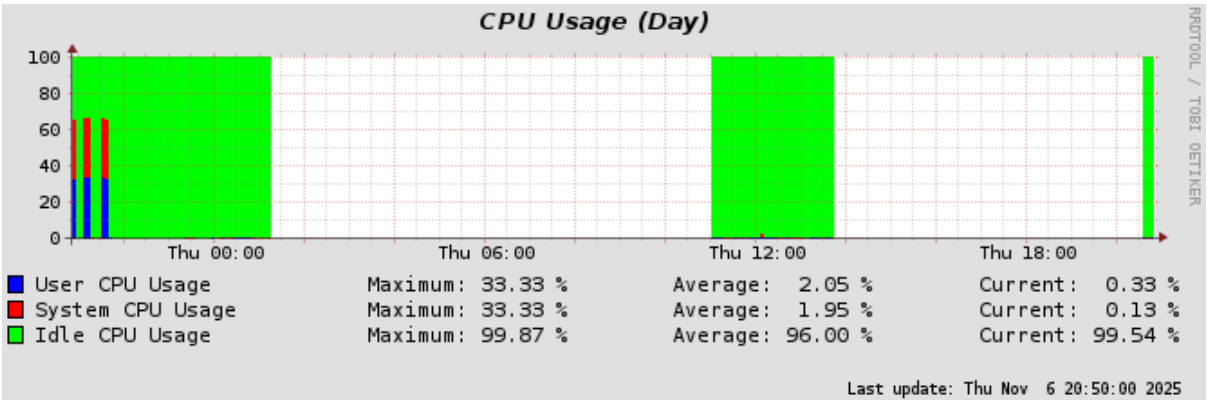


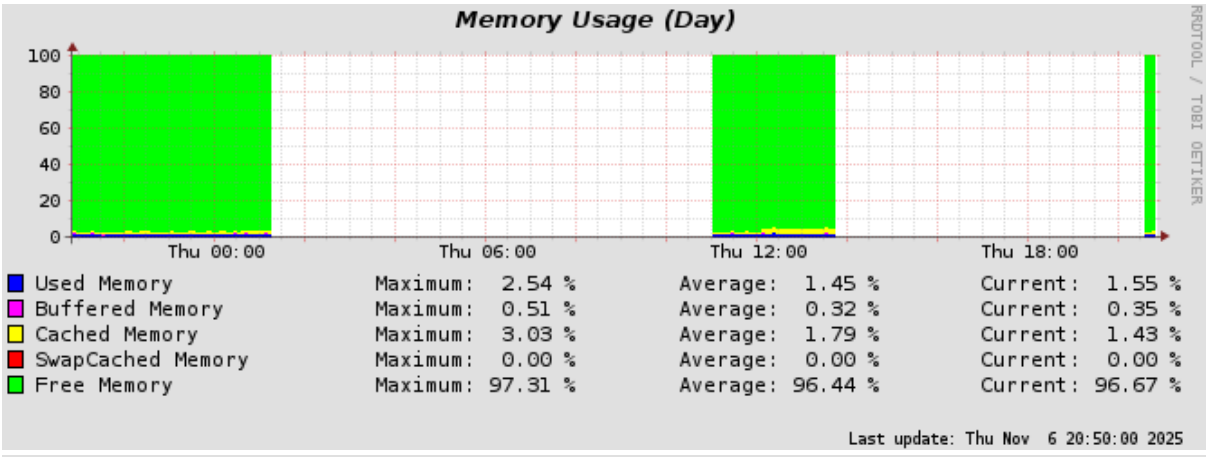
Denial of Service Attacks

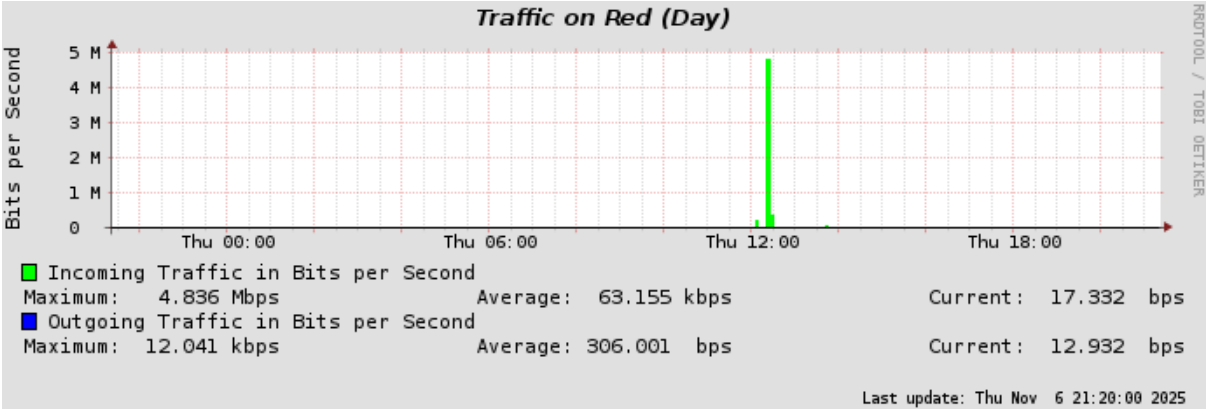
Both ping and fragmented hping3 requests to IPCop are successful, however they use minimal resources and not enough traffic to overwhelm the firewall.



```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 -f -d 1500 192.168.56.101
HPING 192.168.56.101 (eth0 192.168.56.101): icmp mode set, 28 headers + 1500 d
ata bytes
len=1500 ip=192.168.56.101 ttl=64 DF id=24197 icmp_seq=0 rtt=8.2 ms
len=1500 ip=192.168.56.101 ttl=64 DF id=24291 icmp_seq=1 rtt=6.5 ms
len=1500 ip=192.168.56.101 ttl=64 DF id=24363 icmp_seq=2 rtt=19.8 ms
len=1500 ip=192.168.56.101 ttl=64 DF id=24405 icmp_seq=3 rtt=20.8 ms
len=1500 ip=192.168.56.101 ttl=64 DF id=24472 icmp_seq=4 rtt=13.8 ms
len=1500 ip=192.168.56.101 ttl=64 DF id=24566 icmp_seq=5 rtt=10.4 ms
len=1500 ip=192.168.56.101 ttl=64 DF id=24601 icmp_seq=6 rtt=12.5 ms
^C
── 192.168.56.101 hping statistic ──
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 6.5/13.1/20.8 ms
```

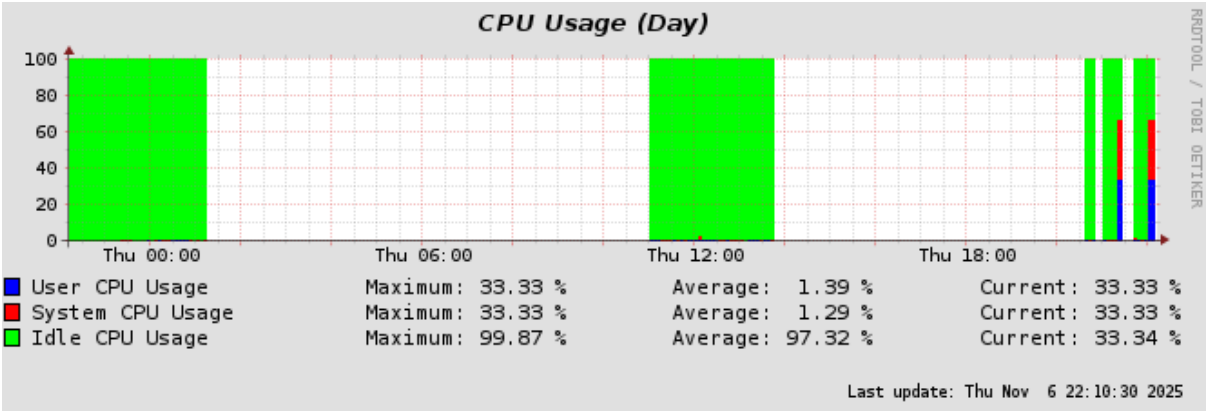


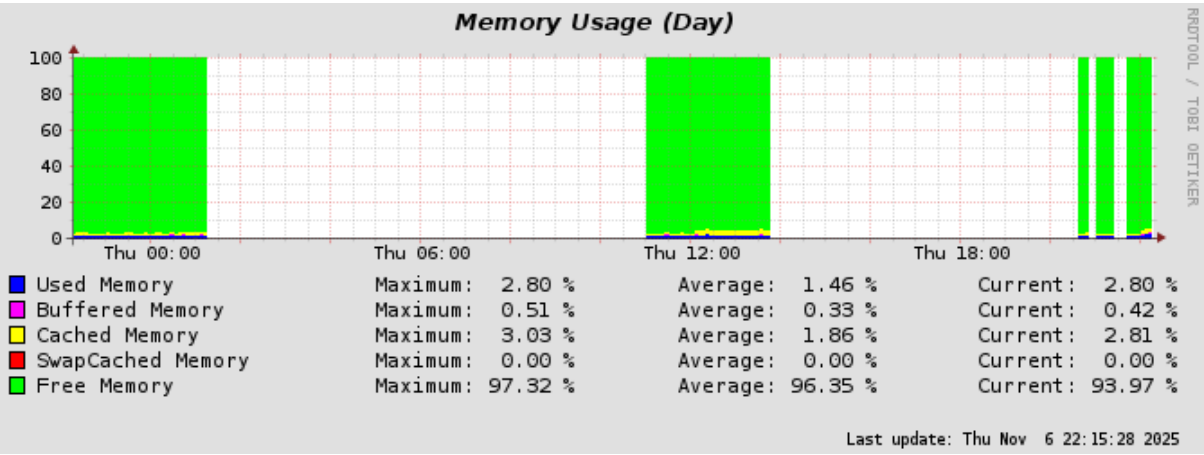


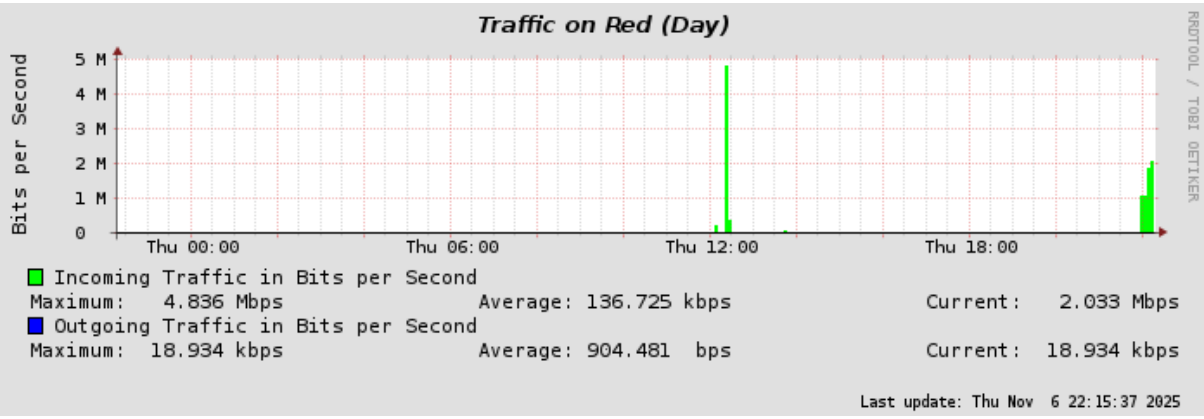


Flooding IPCop with hping3 requests however does use up to two thirds of the CPU and 2.8% of the memory and generates around 2.033Mbps of incoming traffic and around 18.934kbps of outgoing traffic after 20 minutes.

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 --flood -f -d 1500 192.168.56.101
HPING 192.168.56.101 (eth0 192.168.56.101): icmp mode set, 28 headers + 1500 data bytes
hping in flood mode, no replies will be shown
^C
── 192.168.56.101 hping statistic ──
42803 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

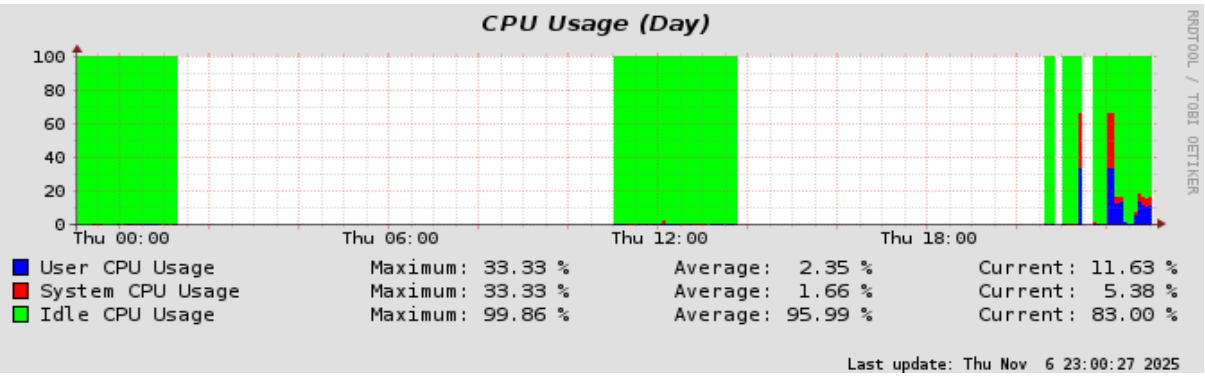


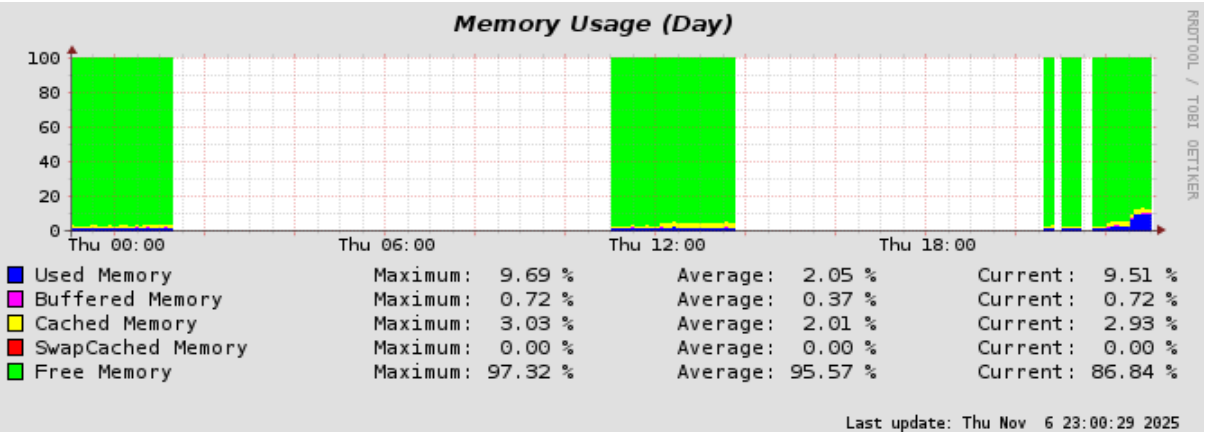


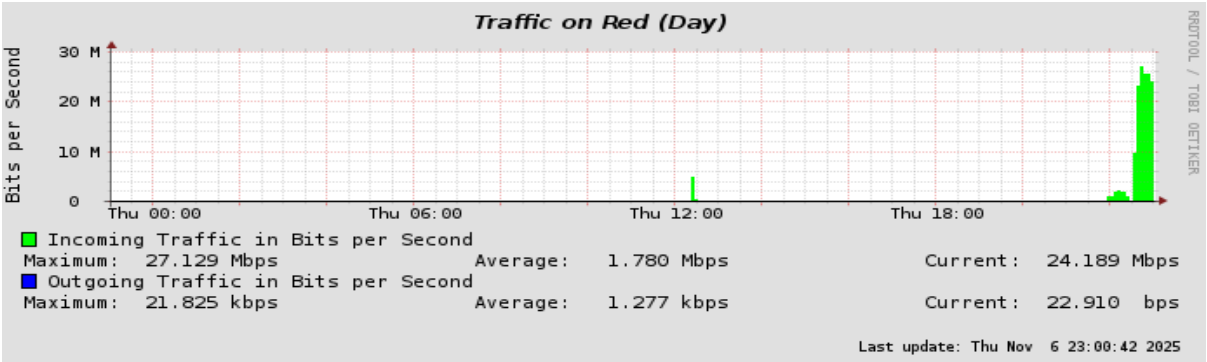


Using random source IP addresses for a flood hping3 attack only used 17% of the CPU but 9.51% of the memory and generates around 24.189Mbps of incoming traffic and around 22.910bps of outgoing traffic after 20 minutes.

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 --flood --rand-source -x -d 1500 192.168.56.101
HPING 192.168.56.101 (eth0 192.168.56.101): icmp mode set, 28 headers + 1500 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.56.101 hping statistic ---
87920 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

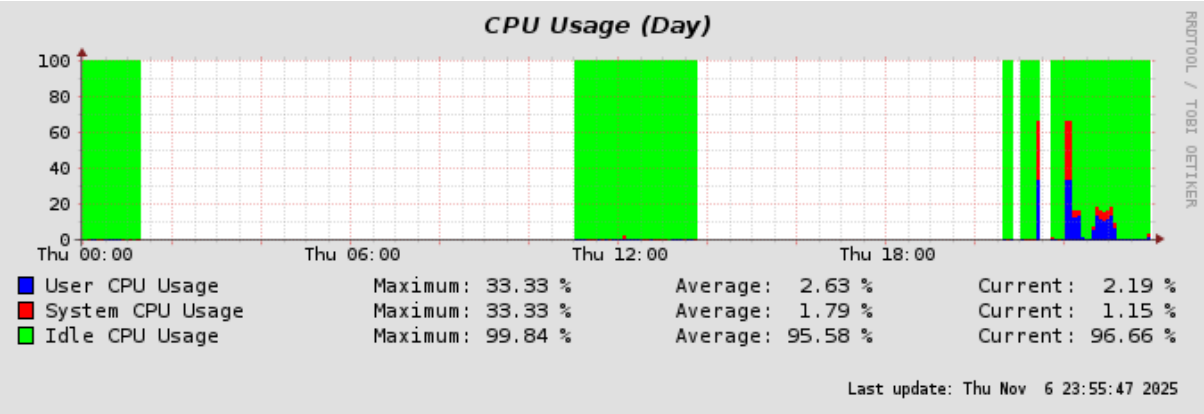


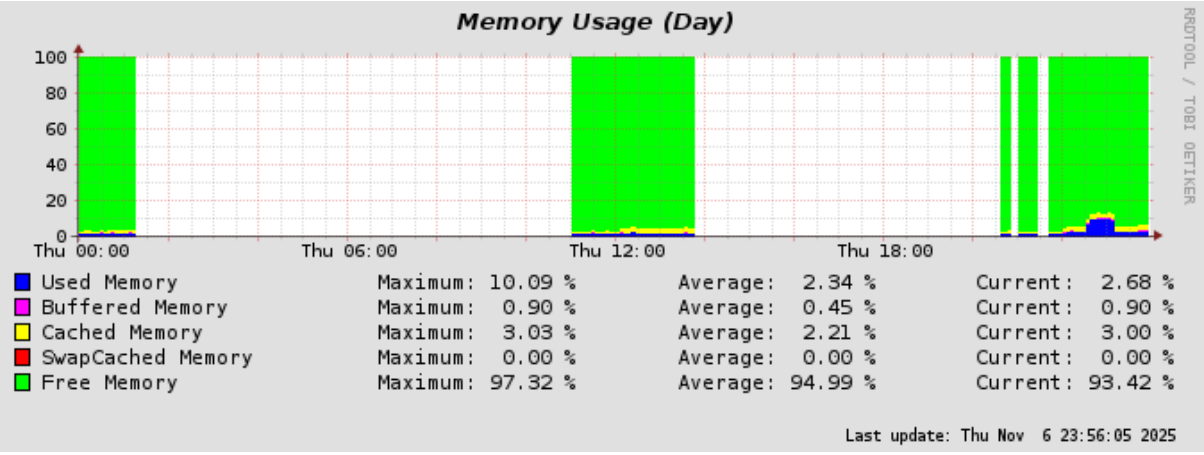


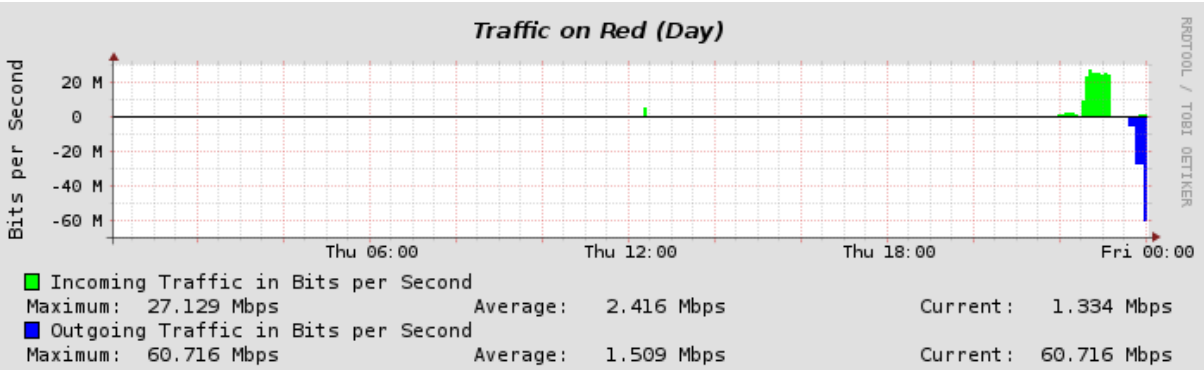


Another DoS attack is done using wrk which generated 190.84MB of HTTP traffic from 18315 requests in twenty seconds. As a result the requests used 3.34% of the CPU and 2.68% of the memory and generates around 1.334Mbps of incoming traffic and around 60.716Mbps of outgoing traffic.

```
┌──(kali㉿kali)-[~]
└─$ wrk -t 4 -c 4 -d 20 http://192.168.56.101
Running 20s test @ http://192.168.56.101
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency     4.36ms    0.96ms  31.60ms   86.53%
    Req/Sec   229.10     20.70   300.00     75.50%
  18315 requests in 20.09s, 190.84MB read
Requests/sec:    911.45
Transfer/sec:      9.50MB
```

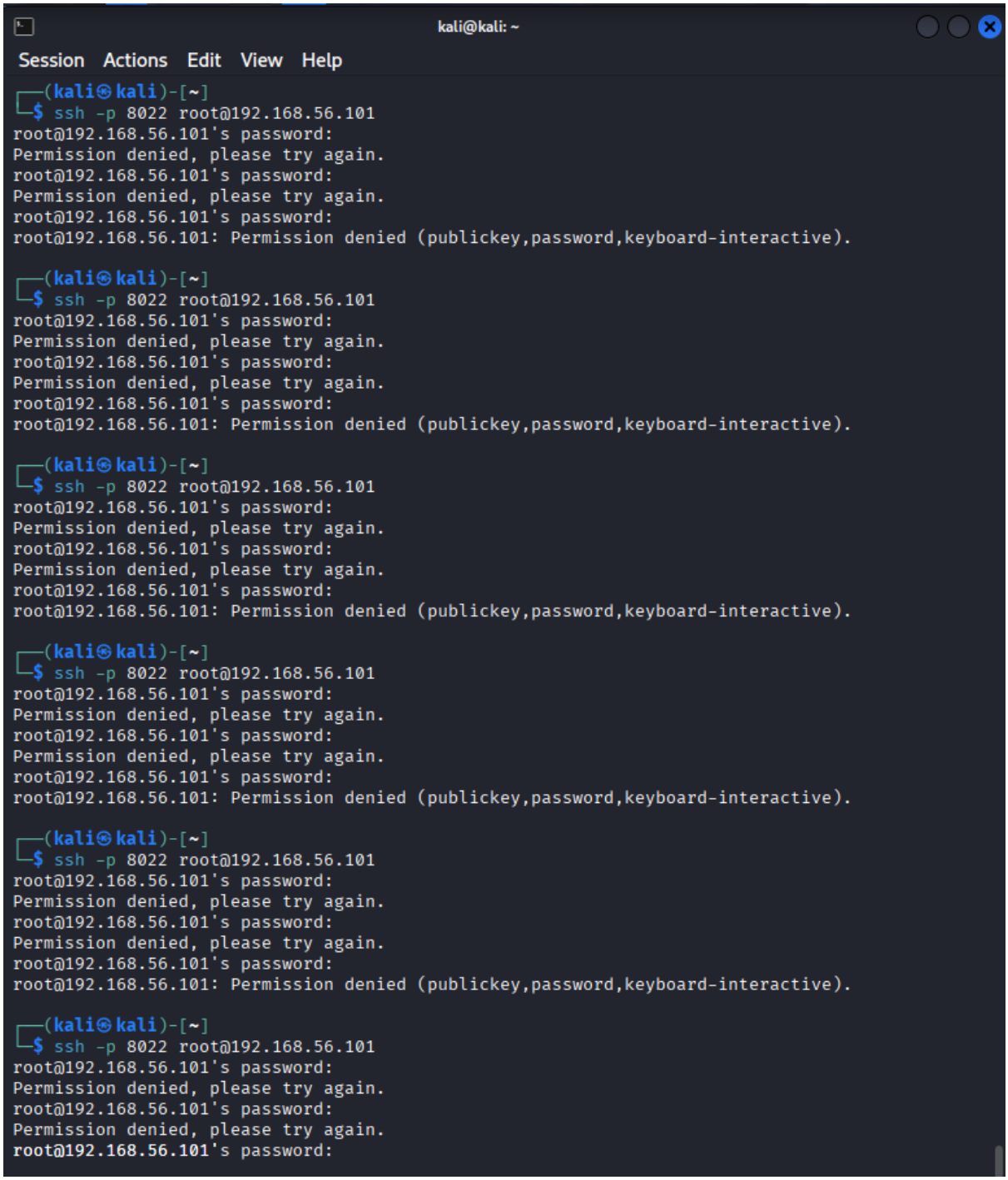




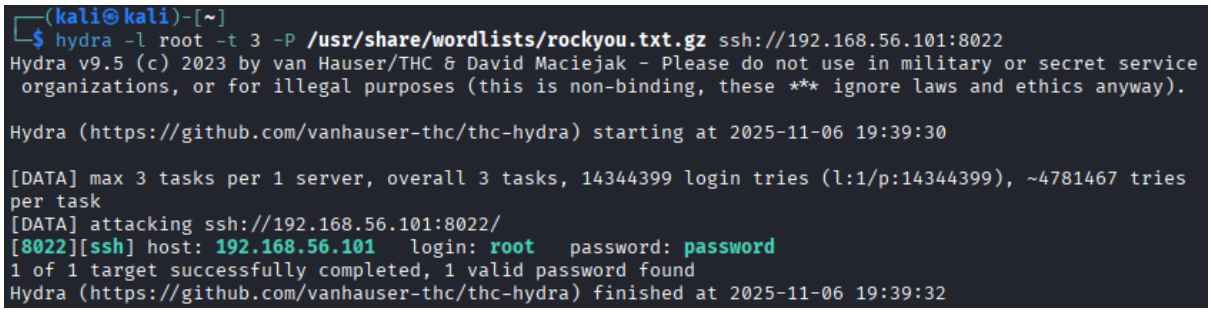


SSH Attacks

Anyone can attempt to log in to the firewall using an SSH connection. As IPCop does not have a lockout policy this allows attackers to use brute force attack.



Brute Force attempts using a list of 14 million leaked RockYou passwords are successful for basic passwords configured for root on IPCop:



```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.101:8022
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
 organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-06 19:47:44
[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p:14344399), ~4781467 tries
per task
[DATA] attacking ssh://192.168.56.101:8022/
[STATUS] 3947.00 tries/min, 3947 tries in 00:01h, 14340452 to do in 60:34h, 3 active
[8022][ssh] host: 192.168.56.101   login: root   password: dublin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-06 19:48:48
```

Reconfiguration

IPCop no longer used snort, an Intrusion Detection and Prevention System, in its last version. Instead, a firewall to drop ICMP packets coming from the red network has been added with the aim to block ICMP flood attacks using ping or hping3.

External IPCop Access:

#	Net Iface	Source		Destination
1	Red	Any	IPCop	IPCop : Ping

By default, HTTP traffic goes into the Firewall's Traffic Shaping Medium priority group which has a guaranteed bandwidth of 607Kbit while the Low priority group has a smaller guaranteed bandwidth of 405Kbit.

```
root@ipcop:~ # tc -s -d class show dev wan-1 | head -17
class htb 1:1 root rate 1013Kbit ceil 1013Kbit linklayer ethernet burst 2865b/8
mpu 0b overhead 0b cburst 2865b/8 mpu 0b overhead 0b level 7
 Sent 84 bytes 2 pkt (dropped 0, overlimits 0 requeues 0)
 rate 16bit 0pps backlog 0b 0p requeues 0
 lended: 0 borrowed: 0 giants: 0
 tokens: 347719 ctokens: 347719

class htb 1:10 parent 1:1 leaf 10: prio 1 quantum 1582 rate 810Kbit ceil 1013Kbi
t linklayer ethernet burst 2611b/8 mpu 0b overhead 0b cburst 2865b/8 mpu 0b over
head 0b level 0
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 rate 0bit 0pps backlog 0b 0p requeues 0
 lended: 0 borrowed: 0 giants: 0
 tokens: 403078 ctokens: 353640

class htb 1:20 parent 1:1 leaf 20: prio 2 quantum 1185 rate 607Kbit ceil 1013Kbi
t linklayer ethernet burst 2357b/8 mpu 0b overhead 0b cburst 2865b/8 mpu 0b over
head 0b level 0
 Sent 84 bytes 2 pkt (dropped 0, overlimits 0 requeues 0)
 rate 16bit 0pps backlog 0b 0p requeues 0
 lended: 2 borrowed: 0 giants: 0
 tokens: 475703 ctokens: 347719
```

```
root@ipcop:~ # tc -s -d class show dev wan-1 | tail -6
class htb 1:30 parent 1:1 leaf 30: prio 3 quantum 1000 rate 405Kbit ceil 1013Kbi
t linklayer ethernet burst 2106b/8 mpu 0b overhead 0b cburst 2865b/8 mpu 0b over
head 0b level 0
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 rate 0bit 0pps backlog 0b 0p requeues 0
 lended: 0 borrowed: 0 giants: 0
 tokens: 650000 ctokens: 353640
```

To minimise the impact of DoS attacks, HTTP traffic has been configured to the Low priority group.

Traffic shaping services

Priority	Port	Protocol	Action
Low	80	tcp	

SSH access to IPCop from untrusted sources in the red network can be prevented by

removing the firewall rule that allowed access as the web interface does not have a feature to limit the number of requests before a timeout, or to allow connections only during certain hours or days. It is also not possible to have more than one source network or address per rule. It is not necessary to add a new rule that explicitly denies SSH access from the red network as all packets from the red network are already dropped by default, but one has been added anyway for clarity.

External IPCop Access:

#	Net Iface	Source			Destination
1	Red	Any		IPCop	IPCop : IPCopSSH

Attacks after firewall reconfiguration

Reconnaissance Attacks

Nmap scan shows less information than before. This time it only shows one open port as 80/TCP since the SSH port 8022/TCP has been blocked. The Operating System detector still identifies that IPCop is a Linux but no longer guesses that it is a firewall and nor that the distribution is IPFire.

```
┌──(kali㉿kali)-[~]
└─$ nmap -A -p- 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 08:37 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns
or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.0013s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 08:00:27:90:FF:8C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose|router
Running (JUST GUESSING): Linux 4.X|5.X (91%), MikroTik RouterOS 7.X (85%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:
linux_kernel:5.6.3
Aggressive OS guesses: Linux 4.19 - 5.15 (91%), Linux 4.15 - 5.19 (85%), Linux 4.19 (85%), Linux 5.0 - 5
.14 (85%), OpenWrt 21.02 (Linux 5.4) (85%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   1.32 ms 192.168.56.101

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 127.04 seconds
```

Denial of Service Attacks

Ping and hping3 no longer gives a response.

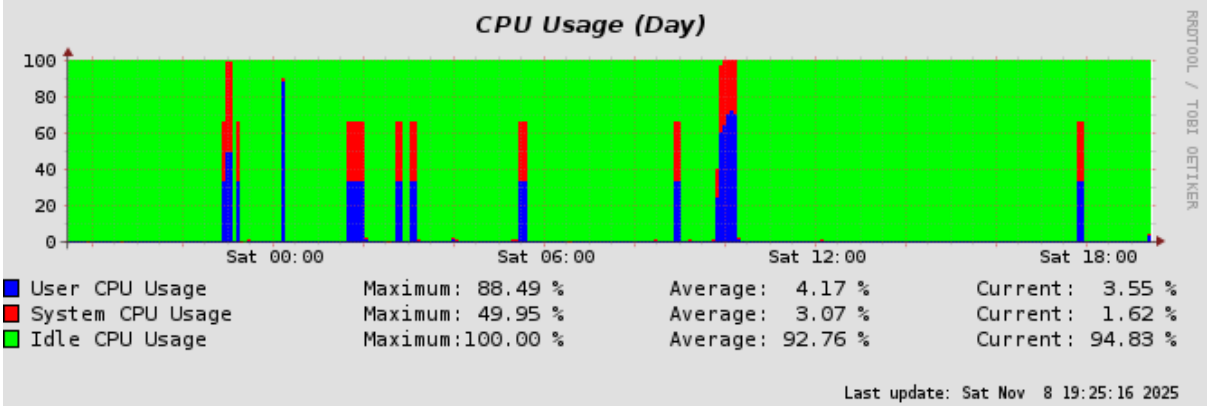
```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.10.101
ping: connect: Network is unreachable
```

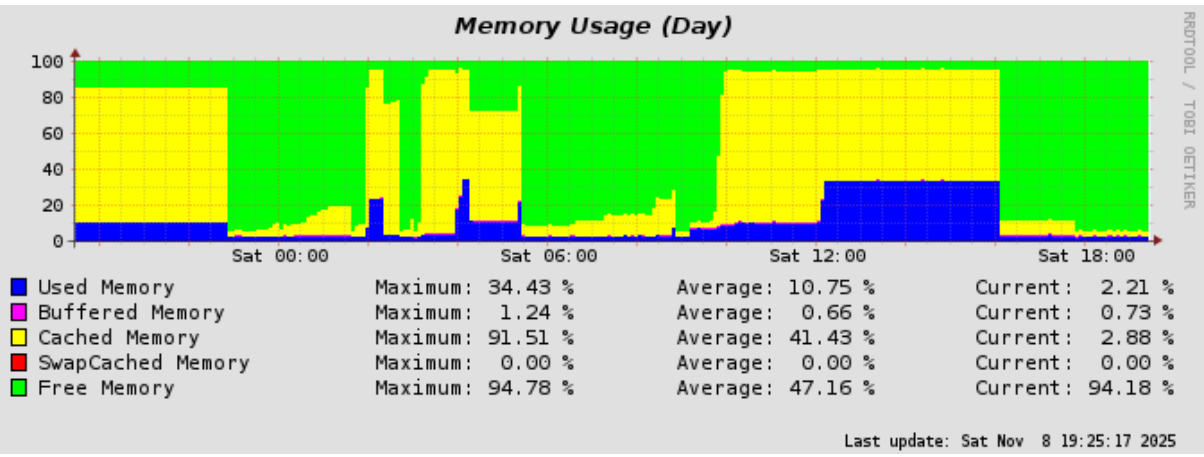
```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 -f -d 1500 192.168.56.101 -c 4
HPING 192.168.56.101 (eth0 192.168.56.101): icmp mode set, 28 headers + 1500 data bytes

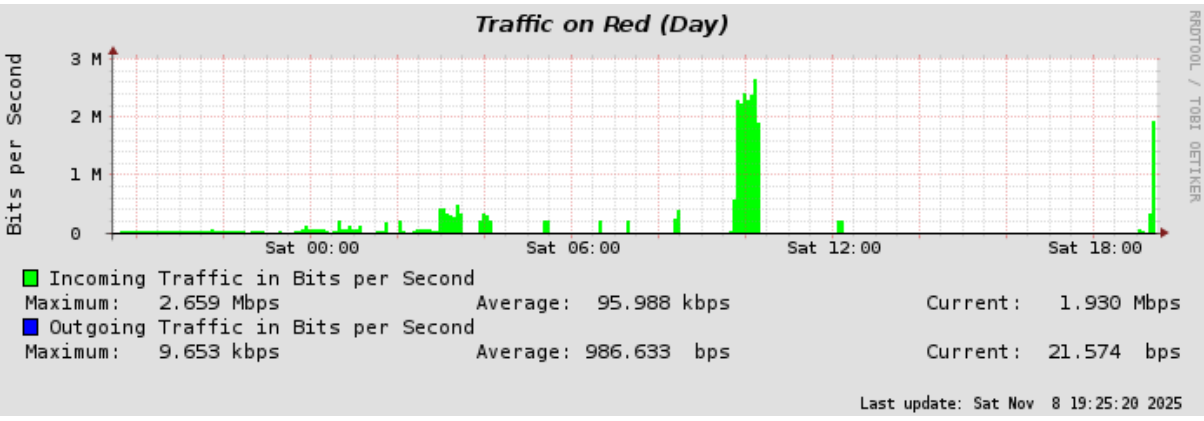
--- 192.168.56.101 hping statistic ---
4 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

Flooding IPCop with hping3 requests for 20 minutes again uses much less CPU (down to 5.17%) and slightly less memory (2.21%). It also generates slightly less incoming traffic of 1.93Mbps and significantly less traffic outgoing traffic of 21.574bps.

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 --flood -f -d 1500 192.168.56.101
HPING 192.168.56.101 (eth0 192.168.56.101): icmp mode set, 28 headers + 1500 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.56.101 hping statistic ---
17593 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

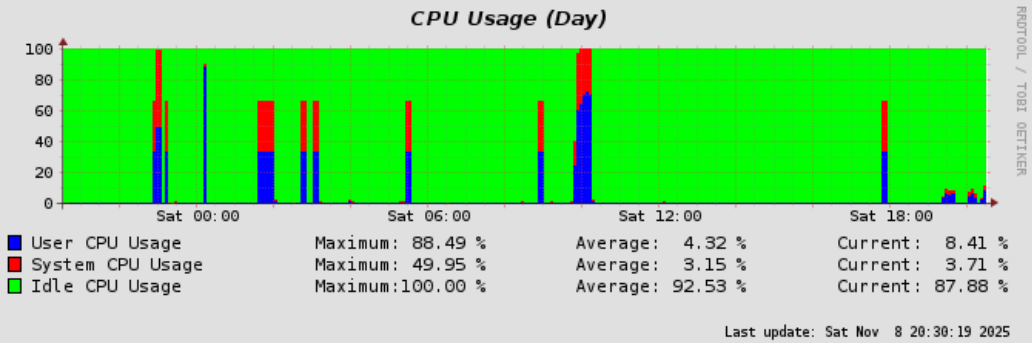


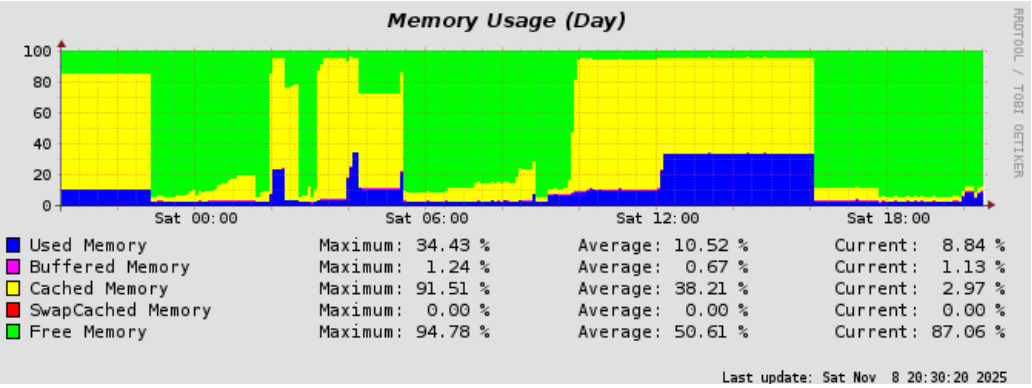


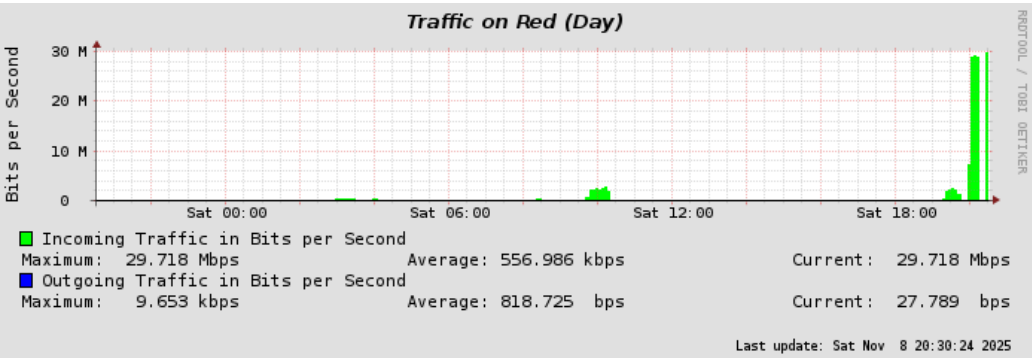


Using random source IP addresses for a flood hping3 attack for 20 minutes attack used less of the CPU (down to 12.12%) and the memory (8.84). It generates more incoming traffic of around 29.718Mbps and outgoing traffic of around 27.789bps.

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 --flood --rand-source -x -d 1500 192.168.56.101
HPING 192.168.56.101 (eth0 192.168.56.101): icmp mode set, 28 headers + 1500 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.56.101 hping statistic ---
343287 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```







The same wrk attack generates a massive decrease down to 2.30MB of HTTP traffic from 220 requests in 20 seconds. It generates around 29.279Kbps of incoming traffic and around 629.675Kbps of outgoing traffic.

```
┌──(kali㉿kali)-[~]
└─$ wrk -t 4 -c 4 -d 20 http://192.168.56.101
Running 20s test @ http://192.168.56.101
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency   363.15ms    9.75ms 396.40ms   89.55%
    Req/Sec     2.40      0.49     3.00     59.55%
  220 requests in 20.07s, 2.30MB read
Requests/sec:     10.96
Transfer/sec:    117.30KB
```

SSH Attacks

Attempts to remotely access IPCop using SSH connections are no longer possible, making brute forcing using the hydra tool unsuccessful. As the default policy and supplementary IPCop rule drops packets, no rejection response is sent.

```
┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
ssh: connect to host 192.168.56.101 port 8022: Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.101:8022
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
 organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-07 07:12:58
[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p:14344399), ~4781467 tries
per task
[DATA] attacking ssh://192.168.56.101:8022/
[ERROR] could not connect to ssh://192.168.56.101:8022 - Timeout connecting to 192.168.56.101
```

IpFire (Lexie Walsh)

Background

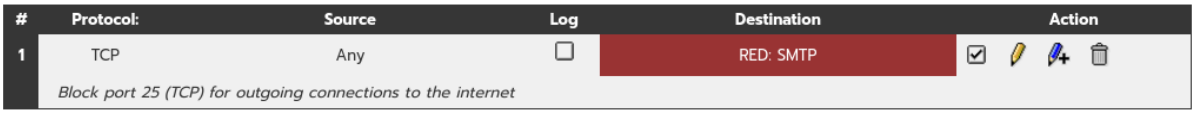
IPFire is an open-source firewall that also uses Iptables. It is based on Linux and offers a web GUI for easier configuration. IpFire uses colours for each network:

- Green for internal network
- Orange for Webservers
- Red for the outside network
- Blue for VPN networks

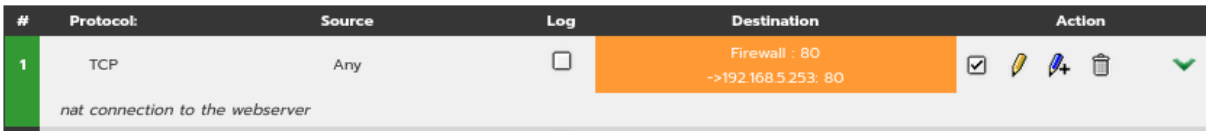
In the case of this project, the green, red and orange networks were configured.

Initial configuration of the firewall

The firewall came with an initial firewall rule to block TCP port 25 (SMTP protocol) for all outgoing internet connections. This port is typically used for communication between mail servers, but it is often exploited as a spam port for malicious traffic.



For the network topology configuration, a port forwarding rule was added to allow HTTP traffic from any network to the web server.



To access the firewall from any network, an SSH rule was added.



Attacks before firewall configuration

All attacks are being done from the WAN (red) network. The red network's IP address is 192.168.56.103.

Reconnaissance Attack

Nmap scan (nmap -A -p- 192.168.56.103)

```
└─$ nmap -A -p- 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-27 12:37 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.103
Host is up (0.00044s latency).
Not shown: 65532 filtered tcp ports (no-response), 1 filtered tcp ports (port
-unreach)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.0 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 08:00:27:05:58:1C (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose|firewall|storage-misc
Running (JUST GUESSING): Linux 4.X|5.X|6.X|2.6.X|3.X (95%), IPFire 2.X (90%),
 QNAP QTS 5.X (90%)
OS CPE: cpe:/o:linux:linux_kernel:4.19 cpe:/o:linux:linux_kernel:5 cpe:/o:ipf
ire:ipfire:2.25 cpe:/o:linux:linux_kernel:6.1 cpe:/o:qnap:qts:5.0 cpe:/o:linu
x:linux_kernel:5.10 cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:3
Aggressive OS guesses: Linux 4.19 (95%), Linux 4.15 (91%), Linux 5.10 - 5.19
(91%), IPFire 2.25 firewall (Linux 4.14) (90%), IPFire 2.27 (Linux 5.15 - 6.1
) (90%), QNAP QTS 5.0 (Linux 5.10) (90%), Linux 2.6.32 - 3.0 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   0.44 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 133.07 seconds
```

This Nmap scan shows:

- That port 22/tcp is open to allow everyone to establish an SSH connection
- That port 80 is open and that the DMZ is up and running
- The type and version of web server used (Apache httpd 2.4.58 (Ubuntu))
- The firewall name, version and what operating system it is based on
- The traceroute to the red network

DoS Attacks

Pings are working

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.56.103 -c 4
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.334 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.390 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.466 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.431 ms

── 192.168.56.103 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3066ms
rtt min/avg/max/mdev = 0.334/0.405/0.466/0.049 ms
```

Hping3:

```
└─$ sudo hping3 -1 -f 192.168.56.103
HPING 192.168.56.103 (eth0 192.168.56.103): icmp mode set, 28 headers + 0 dat
a bytes
len=46 ip=192.168.56.103 ttl=64 id=25898 icmp_seq=0 rtt=10.7 ms
len=46 ip=192.168.56.103 ttl=64 id=26014 icmp_seq=1 rtt=4.8 ms
len=46 ip=192.168.56.103 ttl=64 id=26288 icmp_seq=2 rtt=3.2 ms
len=46 ip=192.168.56.103 ttl=64 id=27106 icmp_seq=3 rtt=2.8 ms
^C
── 192.168.56.103 hping statistic ──
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 2.8/5.4/10.7 ms
```

Hping3 works too

When performing a ping flood using hping3 --flood, the command shows a 100% packet loss, but the firewall is still showing significant outbound traffic. It does the same when using random IP's as source addresses (--rand-source).

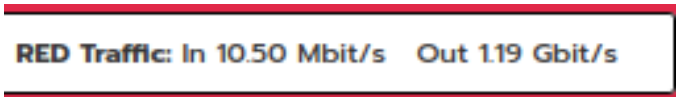
```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 --flood -x -d 1500 192.168.56.103
HPING 192.168.56.103 (eth0 192.168.56.103): icmp mode set, 28 headers + 1500
data bytes
hping in flood mode, no replies will be shown
^C
── 192.168.56.103 hping statistic ──
1168588 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```



Using wrk to put a load on the web server:

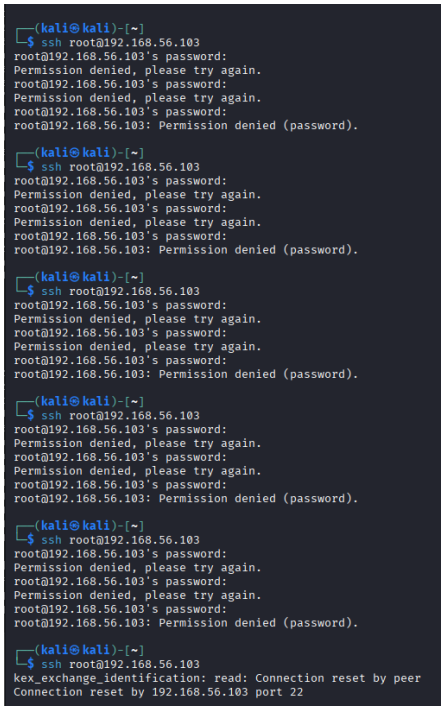
```
┌──(kali㉿kali)-[~]
└─$ wrk -t 4 -c 4 -d 20 http://192.168.56.103
Running 20s test @ http://192.168.56.103
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency   638.90us  670.29us  11.79ms   97.19%
    Req/Sec     1.77k   149.54     2.18k    68.25%
  141237 requests in 20.01s, 1.44GB read
Requests/sec:   7059.13
Transfer/sec:     73.56MB
```

When running wrk for 20 seconds, it puts a lot of strain on the network with 1.2Gbit/s leaving the red network, coming from the orange network.



SSH Attacks

With SSH being accessible from the red network, it is exposed to malicious users, attempting to brute force the password, allowing them to change firewall rules.



For this, an inital attempt was made to establish a connection, and the wrong password was entered. It took 15 failed attempts before the SSH connection was blocked.

Here an example using the brute forcing tool hydra:

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.1
03
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
 military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-27 15:
52:35
[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p
:14344399), ~4781467 tries per task
[DATA] attacking ssh://192.168.56.103:22/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
[INFO] Writing restore file because 2 server scans could not be completed
[ERROR] 1 target was disabled because of too many errors
[ERROR] 1 targets did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-27 15:
52:37
```

Hydra gave an error after the connection reset.

After Firewall Configuration:

- Added IPS rules to prevent basic attacks
- Gave SSH firewall rules for security
- Configured QoS (Quality of Service) for rate-limiting traffic

IPS rules:

The Emergingthreats.net Community rules were used as the ruleset list and enabled on the red, green and orange interfaces. For this project, the rulesets for dos-, drop-, icmp-, icmp_info-, and scan.rules were turned on to prevent the previously tested attacks.

Reconnaissance attack:

```
┌──(kali㉿kali)-[~]
└─$ nmap -A -p- 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-28 15:58 EDT
Nmap scan report for 192.168.56.103
Host is up (0.00099s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 08:00:27:05:58:1C (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose|router|firewall
Running (JUST GUESSING): Linux 4.X|5.X|2.6.X|3.X (97%), MikroTik RouterOS 7.X
 (89%), IPFire 2.X (87%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:
linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3.10 cpe:/o:linux:linux_kernel:
6.0 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3 cpe:/o:ipfire:
ipfire:2.27
Aggressive OS guesses: Linux 4.19 - 5.15 (97%), Linux 4.15 - 5.19 (91%), Linu
x 5.0 - 5.14 (91%), OpenWrt 21.02 (Linux 5.4) (91%), Linux 2.6.32 (90%), Linu
x 2.6.32 or 3.10 (90%), Linux 4.19 (89%), Linux 6.0 (89%), MikroTik RouterOS
7.2 - 7.5 (Linux 5.6.3) (89%), Linux 4.0 - 4.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   1.00 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 168.84 seconds
```

When performing another Nmap scan after adding IPS, it still finds port 80, but no longer finds port 22. It still recognises that it is a web server, but doesn't show any extra information on it.

OSScan doesn't determine the type of firewall, but correctly identifies it as IPFire and the base operating system as Linux.

DoS Attacks

For ping and hping3 attacks, none of them worked either, with inbound traffic greatly reduced and outbound traffic prevented.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.56.103 -c 4
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.

── 192.168.56.103 ping statistics ──
4 packets transmitted, 0 received, 100% packet loss, time 3050ms
```



Red traffic during the ping flood.

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 -f --flood 192.168.56.10
HPING 192.168.56.10 (eth0 192.168.56.10): icmp mode set, 28 headers + 0 data
bytes
hping in flood mode, no replies will be shown
^C
── 192.168.56.10 hping statistic ──
456819574 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

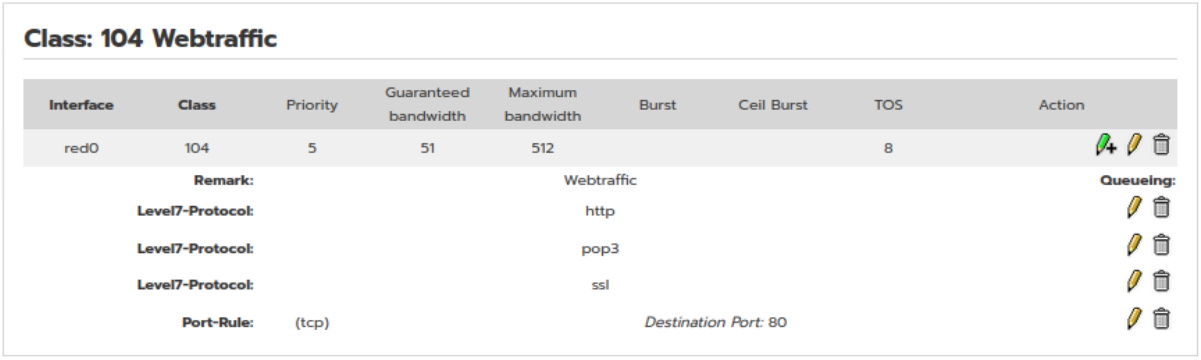
When running hping3 from random sources, the inbound traffic increases due to the random nature of the sources, but the outbound traffic remains constant during the flood.

```
└─$ sudo hping3 -1 -f --flood --rand-source -x 192.168.56.103
```



Traffic during a ping flood.

To prevent a DoS attack from wrk, QoS (Quality of Service) and IPS settings were enabled, which rate-limits types of traffic. A parent class was already created for web traffic after adding firewall rules.



Class: 104 Webtraffic

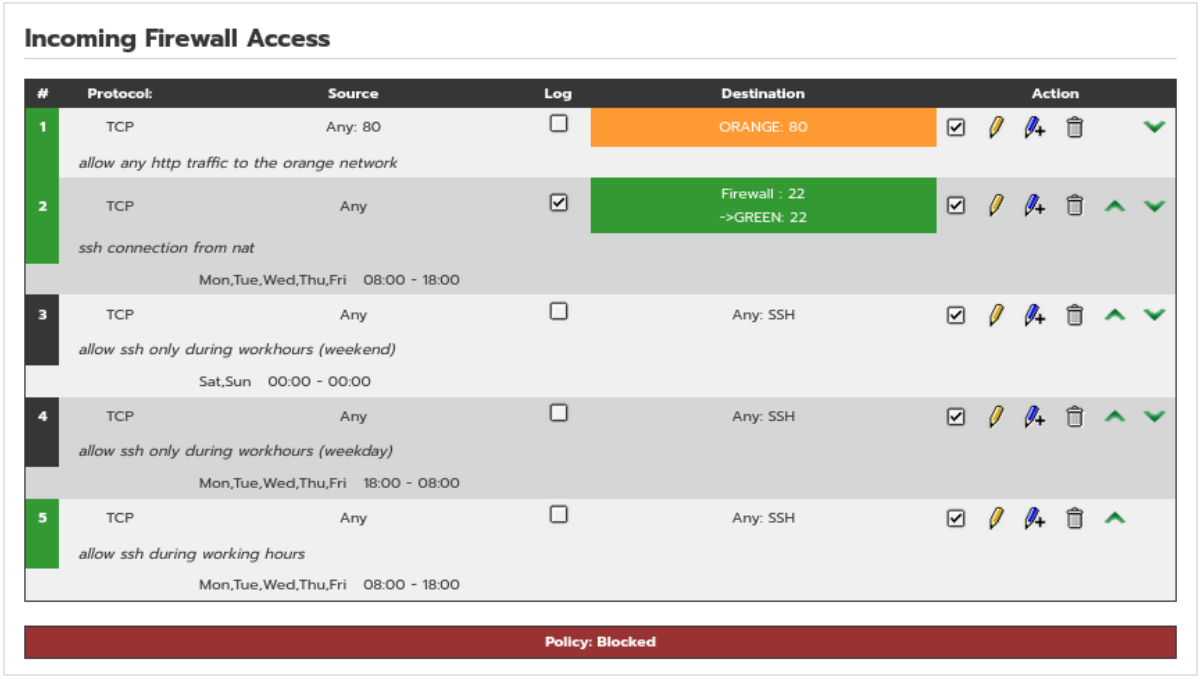
Interface	Class	Priority	Guaranteed bandwidth	Maximum bandwidth	Burst	Ceil Burst	TOS	Action
red0	104	5	51	512			8	
	Remark:			Webtraffic				**Queueing:**
	Level7-Protocol:			http				
	Level7-Protocol:			pop3				
	Level7-Protocol:			ssl				
	Port-Rule:	(tcp)			*Destination Port:* 80			

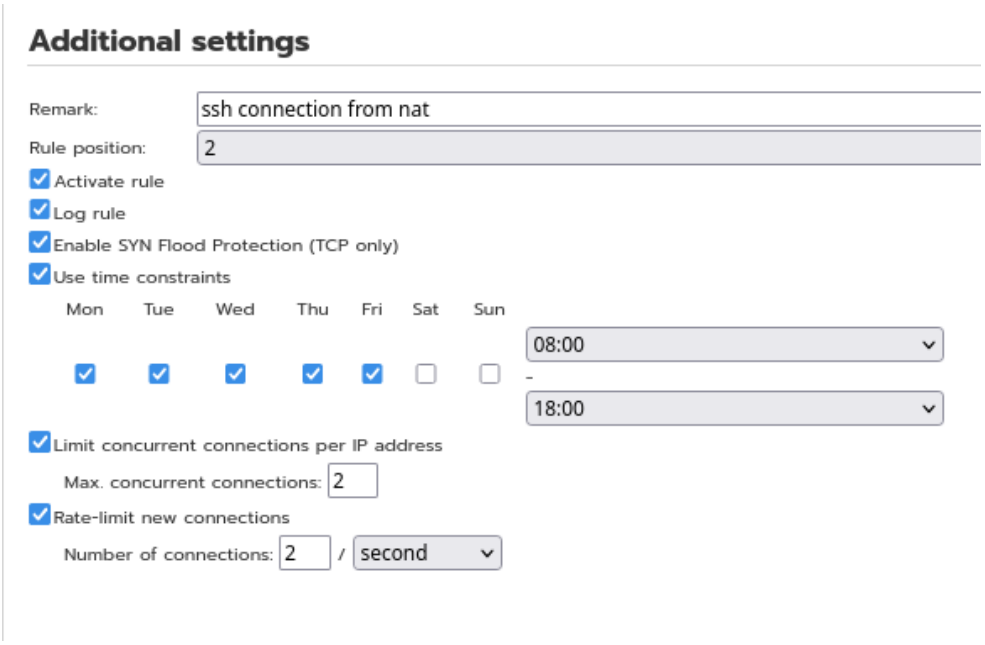
```
┌──(kali㉿kali)-[~]
└─$ wrk -t 4 -c 4 -d 20 http://192.168.56.103
Running 20s test @ http://192.168.56.103
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency   730.56ms   59.19ms 943.06ms   84.26%
    Req/Sec     1.00      0.00     1.00    100.00%
  108 requests in 20.05s, 1.14MB read
Requests/sec:      5.39
Transfer/sec:     58.26KB
```

With QoS turned on and IPS rules in place, the wrk command load decreased drastically

SSH Attacks

Hydra is already being stopped automatically, for extra safely, Time constraints were added that the firewall can only be configured during working hours, limited concurrent connections per IP address and rate limit the total number of connections from new IP addresses. Logging was also turned on to keep track of any suspicious connection requests.





```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.1
03:22
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
 military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-29 13:
33:15
[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p
:14344399), ~4781467 tries per task
[DATA] attacking ssh://192.168.56.103:22/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-29 13:
33:49
```

11:42:48	DROP_INPUT	red0	TCP	192.168.56.101 192.168.56.103	43750 22(SSH)		08:00:27:1f:b7:23
11:42:49	DROP_INPUT	red0	TCP	192.168.56.101 192.168.56.103	43750 22(SSH)		08:00:27:1f:b7:23
11:42:50	DROP_INPUT	red0	TCP	192.168.56.101 192.168.56.103	43750 22(SSH)		08:00:27:1f:b7:23
11:42:51	DROP_INPUT	red0	TCP	192.168.56.101 192.168.56.103	43750 22(SSH)		08:00:27:1f:b7:23

Comparison Tables

Reconnaissance Attacks

Firewall	Blocked
OPNSense	**Partially mitigated:** Nmap shows lots of information in beginning but can be reduced slightly. IPS/IDS config needed to properly mitigate against it.
IPcop	**Partially mitigated:** Initially showed two open ports, one of which was later closed.
IPfire	**Mitigated:** Shows everything initially and after configuring IPS, gets blocked to only showing open ports

Denial of Service Attacks

ICMP Flood

Firewall	Blocked
OPNSense	**Not mitigated:** Can't be mitigated well, seems to struggle with any rules to limit ICMP. One solution is blocking certain IP ranges but impacts logging and CPU usage.
IPcop	**Partially mitigated:** Less CPU and Memory were used during attacks, but the difference in generated traffic varied.
IPfire	**Mitigated:** ICMP flood was mitigated through IPS.

HTTP Flood

Firewall	Blocked
OPNSense	**Mitigated:** Can mitigate it effectively via shaper.
IPcop	**Mitigated:** Can mitigate by using traffic shaper to categorise in the low priority group.
IPfire	**Mitigated:** Was mitigated using QoS and IPS.

SSH Attacks

Firewall	Blocked
OPNSense	**Mitigated:** No easy way to set specific timeouts (without CLI), however it will still lock user out by default.
IPcop	**Partially mitigated:** IPCop's SSH port closed to the red network as rate limits could not be configured. Brute force attacks do not work but neither do simple SSH connections.
IPfire	**Mitigated:** IpFire automatically times out connections after 15 tries, added time constraints for ssh access from the red network and logging to check for suspicious sign in requests.

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